Dhirubhai Ambani University

DBMS Project

INVESTHUB

Lab Group: 04
Team I'd: 02

Name	Student ID
Jainil Patel	202301205
Gautam Modi	202301214
Priyanshi Gothi	202301226
Jeel Prajapati	202301238

Group-Leader ID: 202301205
Contact No: 9726677113

FD Sets and Normal Froms

1. **User(User_ID, fname, mname, lname, Email, PAN_No, Country, Registered_At, User_Type)**
   a. **FD Set:**
      i. User_ID → {fname, mname, lname, Email, PAN_No, Country, Registered_At, User_Type}
      ii. Email → { PAN_No, fname, mname, lname, Country}
   b. **Minimal FD Set :**
      i. User_ID → { Email, Registered_At, User_Type}
      ii. Email → {PAN_No, fname, mname, lname, Country}
   c. **Key :**
      i. User_ID
   d. **Normal-Form:**
      i. **BCNF:** No (Because in FD(ii) Determinant is not a Key)
      ii. **3NF:** No (Because in FD(ii) Dependent is not a Prime Attribute)
      iii. **2NF:** Yes (Because all the non-prime attributes are (directly or transitively) determined fully by the key.

➢ BCNF Decomposition Algorithm :

1. **User(User_Id, Email, Registered_At, User_Type)**
   a. **FD Set:**
      i. User_ID → { Email, Registered_At, User_Type}
   b. **Key :**
      i. User_ID
   c. **Normal-Form:**
      i. **BCNF :** Yes (Because determinant is the Key)
2. **User_Info(Email, PAN_No, fname, mname, lname, Country)**
   a. **FD Set:**
      i. Email → {PAN_No, fname, mname, lname, Country}
   b. **Key :**

i. Email

c. **<u>Normal-Form:</u>**

i. **BCNF :** Yes (Because determinant is the Key)

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2. Address(<u>User_ID</u>, <u>Address</u>)

a. **<u>FD Set:</u>** There are no non-trivial FDs.

b. **<u>Key:</u>**

i. {User_ID, Address}

c. **<u>Normal-Form:</u>**

i. **<u>BCNF:</u>** Yes (Because this relation is all attribute key relation)

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3. Phone_No(<u>User_ID</u>, <u>Phone_No</u>)

a. **<u>FD Set:</u>** There are no non-trivial FDs.

b. **<u>Key:</u>**

i. {User_ID, Phone_No}

c. **<u>Normal-Form:</u>**

i. **<u>BCNF:</u>** Yes (Because this relation is all attribute key relation)

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4. Analyst(<u>Analyst_ID</u>, Reg_No, Experience_Years)

a. **<u>FD Set:</u>**

i. Analyst_ID → { Reg_No, Experience_Years}

ii. Reg_No → { Analyst_ID, Experience_Years}

b. **<u>Minimal FD Set:</u>**

i. Analyst_ID → { Reg_No, Experience_Years}

ii. Reg_No → Analyst_ID

c. **<u>Keys</u>:**

i. Analyst_ID

ii. Reg_No

d. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

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5. Broker(<u>Broker_ID</u>, Broker_Licence_No, Firm_Name, Experience_Years, Commision_Rate)

a. **<u>FD Set:</u>**
   i. Broker_ID → { Broker_Licence_No, Firm_Name, Experience_Years, Commision_Rate}
   ii. Broker_Licence_No → { Broker_ID, Firm_Name, Experience_Years, Commision_Rate}

b. **<u>Minimal FD Set:</u>**
   i. Broker_ID → { Broker_Licence_No, Firm_Name, Experience_Years, Commision_Rate}
   ii. Broker_Licence_No → Broker_ID

c. **<u>Keys</u> :**
   i. Broker_ID
   ii. Broker_Licence_No

d. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

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6. Investor(<u>Investor_ID</u>, Investor_Type)

a. **<u>FD Set (Minimal):</u>**
   i. Investor_ID → Investor_Type

b. **<u>Key:</u>**
   i. Investor_ID

c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinant is the Key)

---

7. DII(<u>DII_ID</u>, Reg_No, Category, Investment_Sector, Total_Investment)

a. **<u>FD Set:</u>**
   i. DII_ID → { Reg_No, Category, Investment_Sector, Total_Investment}
   ii. Reg_No → { DII_ID, Category, Investment_Sector, Total_Investment}

b. **<u>Minimal FD Set:</u>**
   i. DII_ID → { Reg_No, Category, Investment_Sector, Total_Investment}
   ii. Reg_No → DII_ID

c. **<u>Keys</u>**:
   i. DII_ID
   ii. Reg_No

d. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

---

8. FII(<u>FII_ID</u>, Reg_No, Investment_Sector, Total_Investment)

a. **<u>FD Set:</u>**
   i. FII_ID → { Reg_No, Investment_Sector, Total_Investment}
   ii. Reg_No → { FII_ID, Investment_Sector, Total_Investment}

b. **<u>Minimal FD Set:</u>**
   i. FII_ID → { Reg_No, Investment_Sector, Total_Investment}
   ii. Reg_No → FII_ID

c. **<u>Keys</u>**:
   i. FII_ID
   ii. Reg_No

e. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

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9. Retail_Investor(<u>Investor_ID</u>, Account_Balance,DOB)

a. **<u>FD Set (Minimal):</u>**
   i. Investor_ID → { DOB, Account_Balance}
b. **<u>Key:</u>**
   i. Investor_ID
c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinant is the key)

---

10. Follows(<u>Analyst_ID</u>, <u>Investor_ID</u>)

a. **<u>FD Set (Minimal):</u>** There are no non-trivial FDs.
b. **<u>Key:</u>**
   i. {Analyst_ID, Investor_ID}
c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because This relation is all attribute key relation)

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11. Provides_PlatForm_to(<u>Broker_ID</u>, <u>Investor_ID</u>)

a. **<u>FD Set (Minimal):</u>** There are no non-trivial FDs.
b. **<u>Key:</u>**
   i. {Broker_ID, Investor_ID}
c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because This relation is all attribute key relation)

---

12. Sector(<u>Sector_Name</u>,Sector_Size)

a. **<u>FD Set (Minimal):</u>**
   i. Sector_Name → Sector_Size
b. **<u>Key:</u>**
   i. Sector_Name

c. <u>**Normal-Form:**</u>
    i. **BCNF:** Yes (Because Determinant is the key)

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13. News(<u>Sector_Name</u>, <u>News_Title</u>, <u>Date</u>, Description)

a. <u>**FD Set (Minimal):**</u>
    i. {Sector_Name, News_Title, Date} → Description
b. <u>**Key:**</u>
    i. {Sector_Name, News_Title, Date}
c. <u>**Normal-Form:**</u>
    i. **BCNF:** Yes (Because Determinant is the key)

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14. Company(<u>Company_Name</u>, CEO, Founded_Year, Industry_Name, Headquarter, Total_Shares, NR_QTR, Gross_QTR_Profit, EPS, Sector_Name)

a. <u>**FD Set:**</u>
    i. Company_Name → { CEO, Founded_Year, Industry_Name, Headquarter, Total_Share, NR_QTR, Gross_QTR_Profit, EPS, Sector_Name}
    ii. Industry_Name → Sector_Name
b. <u>**Minimal FD Set:**</u>
    i. Company_Name → { CEO, Founded_Year, Industry_Name, Headquarter, Total_Share, NR_QTR, Gross_QTR_Profit, EPS}
    ii. Industry_Name → Sector_Name
c. <u>**Key:**</u>
    i. Company_Name
d. <u>**Normal-From:**</u>
    i. **BCNF:** No (Because in FD(ii) Determinant is not a Key)

ii. **3NF:** No (Because in FD(ii) Dependent is not a Prime Attribute)
iii. **2NF:** Yes (Because all the non-prime attributes are (directly or transitively) determined fully by the key.

➢ **BCNF Decomposition Algorithm :**

1. **Company(Company_Name, CEO, Founded_Year, Industry_Name, Headquarter, Total_Shares, NR_QTR, Gross_QTR_Profit, EPS)**
   a. **FD Set:**
      i. Company_Name → { CEO, Founded_Year, Industry_Name, Headquarter, Total_Share, NR_QTR, Gross_QTR_Profit, EPS}
   b. **Key:**
      i. Company_Name
   c. **Normal-Form:**
      i. **BCNF:** Yes (Because Determinant is the key)
2. **Industry(Industry_Name, Sector_Name)**
   a. **FD Set:**
      i. Industry_Name → Sector_Name
   b. **Key:**
      i. Industry_Name
   c. **Normal-Form:**
      i. **BCNF:** Yes (Because Determinant is the key)

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15. IPO(Symbol, Company_Name, Open_Date, Close_Date, Issue_Size, Issue_Price, Lot_Size, Exchange, IPO_Type, Status, Listing_Price, Final_GMP)

a. **FD Set:**

i. Symbol → {Company_Name, Open_Date, Close_Date, Issue_Size, Issue_Price, Lot_Size, Exchange, IPO_Type, Status, Listing_Date, Listing_Price, Final_GMP}
ii. Company_Name → {Symbol, Open_Date, Close_Date, Issue_Size, Issue_Price, Lot_Size, Exchange, IPO_Type, Status, Listing_Date, Listing_Price, Final_GMP}

b. **Minimal FD Set:**
   i. Symbol → {Company_Name, Open_Date, Close_Date, Issue_Size, Issue_Price, Lot_Size, Exchange, IPO_Type, Status, Listing_Date, Listing_Price, Final_GMP}
   ii. Company_Name → Symbol
c. **Keys:**
   i. Symbol
   ii. Company_Name
d. **Normal-Form:**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

---

16. Stock(Stock_ID, Exchange, Stock_Name, Current_Price, Company_Name)

a. **FD Set:**
   i. Stock_ID → { Stock_Name, Company_Name}
   ii. Company_Name → { Stock_ID, Stock_Name}
   iii. Stock_Name → {Stock_ID, Company_Name}
   iv. { Stock_ID, Exchange} → { Stock_Name, Current_Price, Company_Name}
   v. { Stock_Name, Exchange} → { Stock_ID, Current_Price, Company_Name}
   vi. { Company_Name, Exchange} → {Stock_ID, Current_Price, Stock_Name}
b. **Minimal FD Set:**
   i. Stock_ID → Stock_Name

ii. Stock_Name → Company_Name
iii. Company_Name → Stock_ID
iv. {Stock_ID, Exchange} → Current_Price
v. {Stock_Name, Exchange} → Current_Price
vi. {Company_Name, Exchange} → Current_Price

c. **<u>Keys:</u>**
   i. {Stock_ID, Exchange}
   ii. {Stock_Name, Exchange}
   iii. {Company_Name, Exchange}

d. **<u>Normal-Form:</u>**
   i. **BCNF:** No (Because in FD(i , ii and iii) determinant is not the key)
   ii. **3NF:** Yes (Because in FD(i, ii and iii) dependent is the Prime attributes, and in FD(iv, v and vi) determinant is the Key)

➢ <u>BCNF Decomposition Algorithm :</u>

1. **Stock_Details(<u>Stock_ID</u>, Stock_Name, Company_Name)**
   a. **<u>FD Set:</u>**
      i. Stock_ID → Stock_Name
      ii. Stock_Name → Company_Name
      iii. Company_Name → Stock_ID
   b. **<u>Keys:</u>**
      i. Stock_ID
      ii. Stock_Name
      iii. Company_Name
   c. **<u>Normal-Form:</u>**
      i. **BCNF:** Yes (Because Determinants in all the FDs are Keys)
2. **Stock_Pricing(<u>Stock_ID</u>, <u>Exchange</u>, Current_Price)**
   a. **<u>FD Set:</u>**
      i. {Stock_ID,Exchange} → Current_Price
   b. **<u>Key:</u>**
      i. {Stock_ID, Exchange}

c. **Normal-Form:**

    i. **BCNF:** Yes (Because Determinant is the key)

Note: We have lost 2 FDs after Decomposition, but there is no effect on the data consistency.

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17 . Dividend(<u>Stock_ID</u>, <u>Exchange</u>, <u>Payment_Date,</u> Payment_Method, Dividend_Amount)

a. **FD Set (Minimal):**

    i. {Stock_ID, Exchange, Payment_Date} → {Dividend_Amount, Payment_Method}

b. **Key:**

    i. {Stock_ID, Exchange, Payment_Date}

c. **Normal-Form:**

    i. **BCNF:** Yes (Because Determinant is the key)

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18. Trend(<u>Stock_ID</u>, <u>Exchange</u>, <u>Date</u>, Open_Price, Close_Price, High_Price, Low_Price, Volume, Year_High, Year_Low, PE_Ratio, Market_Cap)

a. **FD Set (Minimal):**

    i. {Stock_ID, Exchange, Date} → {Open_Price, Close_Price, High_Price, Low_Price, Volume, Year_High, Year_Low, PE_Ratio, Market_Cap}

b. **Key:**

    i. {Stock_ID, Exchange, Date}

c. **Normal-Form:**

    i. **BCNF:** Yes (Because Determinant is the key)

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19. Block_Deals(<u>Deal_ID</u>, Exchange, Deal_Type, Qty_Traded, Investor_ID, Stock_ID, Trade_Price, Deal_Time)

a. **FD Set:**
   i. Deal_ID → {Investor_ID, Stock_ID, Exchange, Deal _Type, Qty_Traded, Trade_Price, Deal_Time}
   ii. {Investor_ID, Stock_ID, Exchange, Deal_Time} → {Deal_ID, Deal_Type, Qty_Traded, Trade_Price}
b. **Minimal FD Set:**
   i. Deal_Id → {Investor_Id, Stock_Id, Exchange, Deal _Type, Qty_Traded, Trade_Price, Deal_Time}
   ii. {Investor_ID, Stock_ID, Exchange, Deal_Time} → {Deal_ID}
c. **Keys:**
   i. Deal_Id
   ii. {Investor_ID, Stock_ID, Exchange, Deal_Time}
d. **Normal-Form:**
   i. **BCNF:** Yes (Because Determinants in all the FDs are Keys)

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20. Stock_Recommandation(Analyst_ID, Investor_ID, Stock_ID, Exchange)

a. **FD Set:** There are no non-trivial FDs.
b. **Key:**
   i. {Analyst_ID, Investor_ID, Stock_ID, Exchange}
c. **Normal-Form:**
   i. **BCNF:** Yes (Because this relation is all attribute key relation)

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21. Watchlist(Investor_ID, Broker_ID, Watchlist_ID, Name, Added_On)

a. **FD Set (Minimal):**
   i. {Investor_ID, Broker_ID, Watchlist_ID} → { Name, Added_On}
b. **Key:**

i. {Investor_ID, Broker_ID, Watchlist_ID}

c. **Normal-Form:**

i. **BCNF:** Yes (Because Determinant is the Key)

---

22. WL_Contains(Investor_ID, Broker_ID, Watchlist_ID, Stock_ID, Exchange)

a. **FD Set:** There are no non-trivial FDs.

b. **Key:**

i. {Investor_ID, Broker_ID, Watchlist_ID, Stock_ID, Exchange}

c. **Normal-From:**

i. **BCNF:** Yes (Because this relation is all attribute key relation)

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23. Portfolio(Investor_ID, Broker_ID, Stock_ID, Exchange, Qty_Held, Avg_Buy_Price, Last_Updated)

a. **FD Set (Minimal):**

i. {Investor_Id, Broker_Id, Stock_Id, Exchange} → {Qty_Held, Avg_Buy_Price, Last_Updated}

b. **Key:**

i. {Investor_ID, Broker_ID, Stock_ID, Exchange}

c. **Normal-Form:**

i. **BCNF:** Yes (Because Determinant is the Key)

---

24. Transaction(Transaction_ID, Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Type, Quantity, Price_At_Transaction, Payment_Method, Transaction_Fee, Status, Transaction_Time)

a. **FD Set:**
    i. Transaction_ID → { Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Type, Quantity, Price_At_Transaction, Payment_Method, Transaction_Fee, Status, Transaction_Time}
    ii. {Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Time} → {Transaction_ID, Transaction_Type, Quantity, Price_At_Transaction, Payment_Method, Transaction_Fee, Status}
b. **Minimal FD Set:**
    i. Transaction_ID → { Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Type, Quantity, Price_At_Transaction, Payment_Method, Transaction_Fee, Status, Transaction_Time}
    ii. {Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Time} → Transaction_ID
c. **Keys:**
    i. Transaction_ID
    ii. {Investor_ID, Broker_ID, Stock_ID, Exchange, Transaction_Time}
d. **Normal-Form:**
    i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

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25. Mutual_Funds(Symbol, Fund_Name, Fund_Manager, Risk_Level, Fund_Type, Expense_Ratio, AUM, NAV, Min_Investment)

a. **FD Set:**
    i. Symbol → { Fund_Name, Fund_Manager, Risk_Level, Fund_Type, Expense_Ratio, AUM, NAV, Min_Investment}
    ii. Fund_Name → { Symbol, Fund_Manager, Risk_Level, Fund_Type, Expense_Ratio, AUM, NAV, Min_Investment}
b. **Minimal FD Set:**

i. Symbol → { Fund_Name, Fund_Manager, Risk_Level, Fund_Type, Expense_Ratio, AUM, NAV, Min_Investment}
ii. Fund_Name → Symbol

c. **<u>Keys:</u>**
   i. Symbol
   ii. Fund_Name

d. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

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26. MF_Carries(<u>MF_Symbol</u>, <u>Stock_ID</u>, <u>Exchange</u>)

a. **<u>FD Set:</u>** There are no non-trivial FDs.

b. **<u>Key:</u>**
   i. {MF_Symbol, Stock_ID, Exchange}

c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because this relation is all attribute key relation)

---

27. MF_Portfolio(<u>Investor_ID</u>, <u>Broker_ID</u>, <u>MF_Symbol</u>, Units_Held, Avg_Buy_NAV, Last_Updated)

a. **<u>FD Set (Minimal):</u>**
   i. {Investor_ID, Broker_Id, MF_Symbol} → { Units_Held, Avg_Buy_NAV, Last_Updated}

b. **<u>Key:</u>**
   i. {Investor_ID, Broker_ID, MF_Symbol}

c. **<u>Normal-Form:</u>**
   i. **BCNF:** Yes (Because Determinant is the Key)

---

28. MF_Transaction(<u>Transaction_ID</u>, Investor_ID, Broker_ID, MF_Symbol, Transaction_Type, Units, Price_At_Transaction, Status, Transaction_Time)

a. **FD Set:**
   i. Transaction_ID → {Investor_ID, Broker_ID, MF_Symbol, Transaction_Type, Units, Price_At_Transaction, Status, Transaction_Time}
   ii. {Investor_ID, Broker_ID, MF_Symbol, Transaction_Time} → {Transaction_ID, Transaction_Type, Units, Price_At_Transaction, Status}

b. **Minimal FD Set:**
   i. Transaction_ID → {Investor_ID, Broker_Id, MF_Symbol, Transaction_Type, Units, Price_At_Transaction, Status, Transaction_Time}
   ii. {Investor_ID, Broker_ID, MF_Symbol, Transaction_Time} → Transaction_ID

c. **Keys:**
   i. Transaction_ID
   ii. {Investor_ID, Broker_ID, MF_Symbol, Transaction_Time}

d. **Normal-Form:**
   i. **BCNF:** Yes (Because Determinants in both the FDs are Keys)

DDL Script

```
CREATE SCHEMA INVESTHUB;
SET SEARCH_PATH TO INVESTHUB;

CREATE TABLE User_Info(
    Email VARCHAR(100) PRIMARY KEY,
    PAN_No CHAR(10) NOT NULL,
    fname VARCHAR(50) NOT NULL,
    mname VARCHAR(50) NOT NULL,
    lname VARCHAR(50) NOT NULL,
    Country VARCHAR(50) NOT NULL
);

CREATE TABLE "User"(
    User_ID INTEGER PRIMARY KEY,
    Email VARCHAR(100) NOT NULL,
    Registered_At DATE NOT NULL,
    User_Type VARCHAR(20) NOT NULL CHECK (User_Type IN
('Investor', 'Broker','Analyst')),
    FOREIGN KEY (Email) REFERENCES User_Info(Email) ON DELETE
CASCADE ON UPDATE CASCADE
);

CREATE TABLE Address(
    User_ID INTEGER,
    "Address" TEXT,
    PRIMARY KEY(User_ID,"Address"),
    FOREIGN KEY(User_ID) REFERENCES "User"(User_ID) ON DELETE
CASCADE
);

CREATE TABLE Phone_No(
```

```
    User_ID INTEGER,
    Phone_No VARCHAR(15),
    PRIMARY KEY(User_ID,Phone_No),
    FOREIGN KEY(User_ID) REFERENCES "User"(User_ID) ON DELETE
CASCADE
);


CREATE TABLE Analyst(
    Analyst_ID INTEGER PRIMARY KEY,
    Reg_No CHAR(12) NOT NULL UNIQUE,
    Experiance INTEGER NOT NULL,
    FOREIGN KEY (Analyst_ID) REFERENCES "User"(User_ID) ON
DELETE CASCADE
);


CREATE TABLE Broker(
    Broker_ID INTEGER PRIMARY KEY,
    -- Official license number of the broker, must be unique
    Broker_Licence_No CHAR(14) NOT NULL UNIQUE,
    Firm_Name VARCHAR(50) NOT NULL,
    Experiance_Years INTEGER NOT NULL,
     -- Standard commission rate charged by the broker
(potentially variable)
    Commision_Rate DECIMAL(3,2),
    FOREIGN KEY(Broker_ID) REFERENCES "User"(User_ID) ON DELETE
CASCADE
);

CREATE TABLE Investor(
    Investor_ID INTEGER PRIMARY KEY,
    Investor_Type VARCHAR(20) NOT NULL CHECK (Investor_Type IN
('FII', 'DII','Retail_Investor')),
```

```
    FOREIGN KEY (Investor_ID) REFERENCES "User"(User_ID) ON
DELETE CASCADE
);

CREATE TABLE Sector(
    Sector_Name VARCHAR(50) PRIMARY KEY,
    Sector_Size INTEGER NOT NULL
);

CREATE TABLE DII(
    DII_ID INTEGER PRIMARY KEY,
    Reg_No CHAR(14) NOT NULL UNIQUE,
    Category VARCHAR(20) NOT NULL,
    Investment_Sector VARCHAR(20) NOT NULL,
    Total_Investment INTEGER NOT NULL,
    FOREIGN KEY (DII_ID) REFERENCES Investor(Investor_ID) ON
DELETE CASCADE,
    FOREIGN KEY (Investment_Sector) REFERENCES
Sector(Sector_Name) ON DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE FII(
    FII_ID INTEGER PRIMARY KEY,
    Reg_No CHAR(14) NOT NULL UNIQUE,
    Investment_Sector VARCHAR(20) NOT NULL,
    Total_Investment INTEGER NOT NULL,
    FOREIGN KEY (FII_ID) REFERENCES Investor(Investor_ID) ON
DELETE CASCADE,
    FOREIGN KEY (Investment_Sector) REFERENCES
Sector(Sector_Name) ON DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE Retail_Investor(
    Investor_ID INTEGER PRIMARY KEY,
```

```
    DOB DATE NOT NULL,
    Account_Balance FLOAT(20) NOT NULL,
    FOREIGN KEY (Investor_ID) REFERENCES Investor(Investor_ID)
ON DELETE CASCADE
);

CREATE TABLE Follows(
    Analyst_ID INTEGER,
    Investor_ID INTEGER,
    PRIMARY KEY(Analyst_ID,Investor_ID),
    FOREIGN KEY (Investor_ID) REFERENCES Investor(Investor_ID)
ON DELETE CASCADE,
    FOREIGN KEY (Analyst_ID) REFERENCES Analyst(Analyst_ID) ON
DELETE CASCADE
);

CREATE TABLE Provides_Platform_To(
    Broker_ID INTEGER,
    Investor_ID INTEGER,
    PRIMARY KEY(Broker_ID,Investor_ID),
    FOREIGN KEY(Broker_ID) REFERENCES Broker(Broker_ID) ON
DELETE CASCADE,
    FOREIGN KEY(Investor_ID) REFERENCES Investor(Investor_ID) ON
DELETE CASCADE
);

CREATE TABLE News(
    Sector_Name VARCHAR(50),
    News_Title VARCHAR(100),
    "Date" DATE ,
    "Description" TEXT NOT NULL,
    PRIMARY KEY(Sector_Name,News_Title,"Date"),
    FOREIGN KEY (Sector_Name) REFERENCES Sector(Sector_Name) ON
DELETE RESTRICT ON UPDATE CASCADE
```

```
);

CREATE TABLE Industry(
    Industry_Name VARCHAR(50) PRIMARY KEY,
    Sector_Name VARCHAR(50) NOT NULL,
    FOREIGN KEY (Sector_Name) REFERENCES Sector(Sector_Name) ON
DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE Company(
    Company_Name VARCHAR(50) PRIMARY KEY,
    CEO VARCHAR(50) NOT NULL,
    Founded_Year CHAR(4) NOT NULL,
    Industry_Name VARCHAR(50) NOT NULL,
    Headquarter VARCHAR(50) NOT NULL,
    Total_Shares BIGINT NOT NULL,
    -- Net Revenue for the latest quarter
    NR_QTR DECIMAL(20,2) NOT NULL,
     -- Gross Profit for the latest quarter
    Gross_QTR_Profit DECIMAL(20,2) NOT NULL,
    -- Earnings Per Share
    EPS FLOAT(10) NOT NULL,
    FOREIGN KEY (Industry_Name) REFERENCES
Industry(Industry_Name) ON DELETE RESTRICT ON UPDATE CASCADE
);


CREATE TABLE IPO(
    Symbol VARCHAR(50) PRIMARY KEY,
    Company_Name VARCHAR(50) NOT NULL UNIQUE,
    Open_Date DATE NOT NULL,
    Close_Date DATE NOT NULL,
    -- Total size of the IPO issue in currency units or shares
    Issue_Size BIGINT NOT NULL,
```

```sql
     -- Price per share for the IPO
    Issue_Price DECIMAL(10,2) NOT NULL,
    -- Minimum number of shares an investor must apply for
    Lot_Size INTEGER NOT NULL,
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    IPO_Type VARCHAR(9) NOT NULL CHECK (IPO_Type IN ('SME',
'MAIN_BOARD')),
    "Status" VARCHAR(10) NOT NULL CHECK ("Status" IN ('Closed',
'Listed','Ongoing','Upcoming')),
    Listing_Date DATE NOT NULL,
    Listing_Price DECIMAL(10,2),
    -- Final Grey Market Premium before listing
    Final_GMP DECIMAL(5,2),
    FOREIGN KEY (Company_Name) REFERENCES Company(Company_Name)
ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE Stock_Info(
    Stock_ID VARCHAR(50) PRIMARY KEY,
    Stock_Name VARCHAR(50),
    Company_Name VARCHAR(50),
    FOREIGN KEY (Company_Name) REFERENCES Company(Company_Name)
ON DELETE CASCADE ON UPDATE CASCADE
);


CREATE TABLE Stock_Pricing(
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    Current_Price DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (Stock_ID,Exchange),
```

```sql
    FOREIGN KEY (Stock_ID) REFERENCES Stock_Info(Stock_ID) ON
DELETE CASCADE ON UPDATE CASCADE
);


CREATE TABLE Dividend(
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    Payment_Date DATE NOT NULL,
    Dividend_Amount FLOAT(7),
    -- Method of payment (e.g., Cash, Stock)
    Payment_Method VARCHAR(50) NOT NULL CHECK (Payment_Method IN
('Cash', 'Stock')),
    PRIMARY KEY(Stock_ID,Exchange,Payment_Date),
    FOREIGN KEY (Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE Trend(
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    "Date" DATE,
    Open_Price DECIMAL(10,2) NOT NULL,
    Close_Price DECIMAL(10,2) NOT NULL,
    High_Price DECIMAL(10,2) NOT NULL,
    Low_Price DECIMAL(10,2) NOT NULL,
     -- Number of shares traded on that date
    Volume BIGINT NOT NULL,
    Year_High DECIMAL(10,2) NOT NULL,
    Year_Low DECIMAL(10,2) NOT NULL,
    -- Price-to-Earnings ratio as of this date
```

```
    PE_Ratio DECIMAL(8,2) NOT NULL,
     -- Market capitalization as of this date (Current_Price * Total_Shares)
    Market_Cap DECIMAL(20,2) NOT NULL,
    PRIMARY KEY (Stock_ID,Exchange,"Date"),
    FOREIGN KEY (Stock_ID,Exchange) REFERENCES Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE CASCADE
);


CREATE TABLE Block_Deals(
    Deal_ID INTEGER PRIMARY KEY,
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE', 'NSE')),
    Deal_Type VARCHAR(4) NOT NULL CHECK (Deal_Type IN ('BUY', 'SELL')),
    Qty_Traded INTEGER NOT NULL,
    Investor_ID INTEGER NOT NULL,
    Stock_ID VARCHAR(50) NOT NULL,
    Trade_Price FLOAT(10) NOT NULL,
    Deal_Time TIMESTAMP NOT NULL,
    UNIQUE(Investor_ID,Stock_ID,Exchange,Deal_Time),
    FOREIGN KEY (Investor_ID) REFERENCES Investor(Investor_ID) ON DELETE CASCADE,
    FOREIGN KEY (Stock_ID,Exchange) REFERENCES Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE CASCADE
);


CREATE TABLE Stock_Recommandation(
    Analyst_ID INTEGER,
    Investor_ID INTEGER,
```

```sql
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) CHECK (Exchange IN ('BSE', 'NSE')),
    PRIMARY KEY(Analyst_ID,Investor_ID,Stock_ID,Exchange),
    FOREIGN KEY (Analyst_ID,Investor_ID) REFERENCES
Follows(Analyst_ID,Investor_ID) ON DELETE CASCADE,
    FOREIGN KEY (Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE Watchlist(
    Investor_ID INTEGER,
    Broker_ID INTEGER,
    Watchlist_ID INTEGER,
    "Name" VARCHAR(50),
    -- Date the watchlist was created
    Added_On DATE NOT NULL,
    PRIMARY KEY (Investor_ID,Broker_ID,Watchlist_ID),
    FOREIGN KEY (Broker_ID,Investor_ID) REFERENCES
Provides_Platform_To(Broker_ID,Investor_ID) ON DELETE CASCADE
);

CREATE TABLE WL_Contains(
    Investor_ID INTEGER,
    Broker_ID INTEGER,
    Watchlist_ID INTEGER,
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    PRIMARY
KEY(Investor_ID,Broker_ID,Watchlist_ID,Stock_ID,Exchange),
    FOREIGN KEY(Investor_ID,Broker_ID,Watchlist_ID) REFERENCES
Watchlist(Investor_ID,Broker_ID,Watchlist_ID) ON DELETE CASCADE,
```

```
    FOREIGN KEY(Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE Portfolio(
    Investor_ID INTEGER,
    Broker_ID INTEGER,
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    Qty_Held INTEGER NOT NULL,
    Avg_Buy_Price DECIMAL(10,2) NOT NULL,
    Last_Updated TIMESTAMP NOT NULL,
    PRIMARY KEY(Investor_ID,Broker_ID,Stock_ID,Exchange),
    FOREIGN KEY(Broker_ID,Investor_ID) REFERENCES
Provides_Platform_To(Broker_ID,Investor_ID) ON DELETE CASCADE,
    FOREIGN KEY(Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE Transaction(
    Transaction_ID INTEGER PRIMARY KEY,
    Investor_ID INTEGER NOT NULL,
    Broker_ID INTEGER NOT NULL,
    Stock_ID VARCHAR(50) NOT NULL,
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    Transaction_Type VARCHAR(4) NOT NULL CHECK (Transaction_Type
IN ('BUY','SELL')),
    Quantity INTEGER NOT NULL,
    Price_At_Transaction DECIMAL(10,2) NOT NULL,
    Payment_Method VARCHAR(20) NOT NULL,
```

```
    Transaction_Fee DECIMAL(10,2) NOT NULL,
    "Status" VARCHAR(10) NOT NULL CHECK ("Status" IN
('Executed', 'Pending', 'Failed', 'Cancelled')),
    Transaction_Time TIMESTAMP NOT NULL,

UNIQUE(Investor_ID,Broker_ID,Transaction_Time,Stock_ID,Exchange)
,
    FOREIGN KEY(Broker_ID,Investor_ID) REFERENCES
Provides_Platform_To(Broker_ID,Investor_ID) ON DELETE CASCADE,
    FOREIGN KEY(Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE Mutual_Funds(
    Symbol VARCHAR(50) PRIMARY KEY,
    Fund_Name VARCHAR(50) NOT NULL,
    Fund_Manager VARCHAR(50) NOT NULL,
    Risk_Level VARCHAR(10) NOT NULL CHECK (Risk_Level IN
('Low','Medium','High')),
    Fund_Type VARCHAR(20) NOT NULL CHECK (Fund_Type IN
('Equity', 'Debt','Hybrid','Index','ETF','Liquid')),
    -- Annual fee charged by the fund (as a percentage)
    Expense_Ratio DECIMAL(5,2) NOT NULL,
    -- Assets Under Management (total value of assets held by
the fund)
    AUM DECIMAL(20,2) NOT NULL,
    -- Net Asset Value per unit of the fund
    NAV DECIMAL(10,2) NOT NULL,
    -- Minimum amount required for initial investment
    Min_Investment DECIMAL(10,2) NOT NULL
);

CREATE TABLE MF_Carries(
```

```sql
    MF_Symbol VARCHAR(50),
    Stock_ID VARCHAR(50),
    Exchange CHAR(3) NOT NULL CHECK (Exchange IN ('BSE',
'NSE')),
    PRIMARY KEY(MF_Symbol,Stock_ID,Exchange),
    FOREIGN KEY(MF_Symbol) REFERENCES Mutual_Funds(Symbol) ON
DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY(Stock_ID,Exchange) REFERENCES
Stock_Pricing(Stock_ID,Exchange) ON DELETE CASCADE ON UPDATE
CASCADE
);

CREATE TABLE MF_Portfolio(
    Investor_ID INTEGER,
    Broker_ID INTEGER,
    MF_Symbol VARCHAR(50),
    -- Number of units of the mutual fund currently held
    Units_Held INTEGER,
     -- Average NAV at which the held units were acquired
    Avg_Buy_NAV DECIMAL(10,2) NOT NULL,
    Last_Updated TIMESTAMP NOT NULL,
    PRIMARY KEY(Investor_ID,Broker_ID,MF_Symbol),
    FOREIGN KEY(MF_Symbol) REFERENCES Mutual_Funds(Symbol) ON
DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY(Broker_ID,Investor_ID) REFERENCES
Provides_Platform_To(Broker_ID,Investor_ID) ON DELETE CASCADE
);

CREATE TABLE MF_Transaction(
    Transaction_ID INTEGER PRIMARY KEY,
    Investor_ID INTEGER NOT NULL,
    Broker_ID INTEGER NOT NULL,
    MF_Symbol VARCHAR(50) NOT NULL,
```

```
    Transaction_Type VARCHAR(4) NOT NULL CHECK (Transaction_Type
IN ('BUY', 'SELL', 'SIP')),
    Units INTEGER NOT NULL,
    Price_At_Transaction DECIMAL(10,2) NOT NULL,
    "Status" VARCHAR(10) NOT NULL CHECK ("Status" IN
('Processed', 'Pending', 'Failed', 'Cancelled')),
    Transaction_Time TIMESTAMP NOT NULL,
    UNIQUE(Investor_ID,Broker_ID,Transaction_Time,MF_Symbol),
    FOREIGN KEY(Broker_ID,Investor_ID) REFERENCES
Provides_Platform_To(Broker_ID,Investor_ID) ON DELETE CASCADE,
    FOREIGN KEY(MF_Symbol) REFERENCES Mutual_Funds(Symbol) ON
DELETE CASCADE ON UPDATE CASCADE
);
```